

Dexter Fernandes

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Machine Learning Engineer

Machine Learning engineer, 3+ years building production image processing applications in modern C++17 and Python. Rebuilt an OpenVINO inference path in C++17 on Intel VPU, 5 to 15 FPS, and wrote a CUDA C++ TensorRT plugin. I ship ML pipelines from training to quantised edge models running 24/7 unattended: crashes down 67%, false positives down 33%. Comfortable debugging across the hardware-software boundary. MSc Robotics, Bristol.

WORK EXPERIENCE

GrowthStage • Remote

05/2026 - Present

Technical Assessor, Computer Vision and Software Engineering

- Evaluated computer vision and SE candidates on their technical depth across SE practices, model deployment, and MLOps practices, maintaining direct exposure to current production engineering standards.

Esbaa • Muscat, Oman

01/2023 - 08/2023

AI Engineer

- Owned a real-time NVIDIA DeepStream 6.x video inference pipeline running 24/7 on BP rigs, from model training and convergence debugging to production deployment, raising F1 15% via adversarial-condition training using PyTorch.
- Built OpenCV, Pillow and scikit-image preprocessing to detect mud and sand contamination on camera feeds, cutting false positives 13% in harsh industrial conditions.
- Hardened the pipeline with Dockerised microservices, Kafka, structured logging, health checks and safe restarts, cutting live crashes 67% on unattended edge devices.
- Developed benchmarking and profiling harnesses to evaluate open-source models (YOLOv5, EfficientDet) using PyTorch and torchvision, quantifying accuracy, latency and stability pre- and post-optimisation.
- Applied Bayesian optimisation, ablation studies, hypothesis testing and confidence intervals in Jupyter, tracking experiments in Weights & Biases to support model selection, threshold tuning and failure analysis.
- Delivered in Agile/SCRUM sprints with project managers and rig operators, documenting failure modes and recovery procedures and briefing leadership on results.
- Led peer code reviews in Git and introduced pytest and unittest coverage with linting into CI/CD validation workflows for edge inference modules, improving release consistency across hardware targets.

LivNSense Technologies • Bengaluru, India

05/2020 - 12/2022

Lead Software Engineer - AI

- Led the development of a two-stage person detection and helmet-colour classification pipeline in TensorFlow and Keras, cutting false positives 33% for Austin Industries.
- Rebuilt the inference path in C++17 with OpenVINO on Intel VPU using object-oriented design and asynchronous multi-processing, raising throughput from 5 to 15 FPS within fixed memory and latency budgets.
- Compressed models 50% through quantisation-aware training, structured pruning and layer fusion in ONNX and OpenVINO while holding accuracy targets.
- Deployed containerised ML inference microservices on Azure with REST endpoints and queue-based async processing, owning autoscaling, monitoring and cost-per-inference.
- Wrote gtest unit tests and documented interfaces for the C++ inference path, catching regressions before deployment.
- Designed the PostgreSQL schema routing real-time incidents into downstream analytics, writing the joins and CTEs behind operational reporting.
- Automated labelling and dataset versioning, cutting manual labelling effort by 80% and reducing model rollout time from 72 hours to under 24.

PROJECTS

DeepStream RTSP Pipeline

06/2026

- Built a multi-stream NVIDIA DeepStream 9.0 / GStreamer RTSP video analytics pipeline running YOLO detection with NvDCF tracking, metadata extraction, anonymisation and re-streamed output across 3 concurrent camera feeds.
- Optimised a YOLO26n TensorRT FP16 detector on 6GB Turing hardware using dynamic-batch engine conversion and p99 latency profiling, sustaining 29.7 FPS per stream live and 131.3 FPS unthrottled, validated FP16 against FP32 for accuracy.
- Implemented a C++/CUDA TensorRT plugin (IPluginV2DynamicExt) moving post-processing into the inference engine.
- Hardened with GPU smoke tests, model-promotion gates, structured JSON logging and per-camera health metrics covering liveness, dropped frames and reconnect events.

Modern Edge CV Benchmarking

03/2026

- Benchmarked YOLOv8, YOLOv10, RT-DETR, EfficientViT, MobileSAM and Depth Anything v2 on edge hardware, comparing accuracy, latency and memory under matched conditions.
- Built a repeatable harness to hold input resolution, batch size and precision constant across architectures, isolating model choice from deployment configuration.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

09/2023 - 09/2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system using RTAB-Map on a Clearpath Husky, integrating LiDAR and radiological data for localisation and mapping in inspection scenarios.
- Validated sensor placement for coverage and calibrated LiDAR extrinsics to the robot base, including the Velodyne VLP-16 and Intel RealSense D435i.
- Implemented a ROS2 (rclcpp) point cloud pipeline in C++17 using PCL, with tf2-based frame management and message synchronisation to resolve transform timing issues.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

07/2016 - 05/2020

- Built a foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

SKILLS

- **Computer Vision & Deep Learning:** ByteTrack, Data augmentation, EfficientDet, HuggingFace, Image classification, Keras, NumPy, NvDCF, NvSort, Object detection, Object tracking, OpenCV, Pillow, PyTorch, RT-DETR, Scikit-image, Segmentation, TensorFlow, Torchvision, YOLO
- **Edge Inference and Optimisation:** custom TensorRT plugins, FP16/INT8 Quantisation, GStreamer, Intel Movidius Myriad VPU, NVIDIA DeepStream, NVIDIA Jetson AGX Xavier, ONNX, OpenVINO, TensorRT
- **Programmng, Cloud & MLOps:** Apache Airflow, Apache Kafka, Bash, C++17, CI/CD, CMake, CUDA C++, Docker, DVC (Data Version Control), Git, gtest, Jupyter, Jupyter, Linux, Microsoft Azure, MLflow, Multi-processing, Optuna, PIL (Pillow), pytest, Python, REST APIs, SQL (PostgreSQL), unittest, wandb
- **Professional Skills:** Jira
- **Annotation & Interpretability:** Confusion-matrix-driven failure clustering, CVAT, GRAD-CAM, TIDE (Toolbox for Identifying Detection Errors)